

SENG 430 — Software Quality Assurance

Code Documentation Coverage Report

HEALTH-AI ML Learning Tool · Sprint 5

Project	Scope	Generated	Sprint
HEALTH-AI	frontend/js/*.js	2026-04-28 12:17	Sprint 5

1. Executive Summary

This report verifies function-level code documentation coverage based on JSDoc blocks placed immediately above function declarations and expressions within the HEALTH-AI ML Learning Tool codebase.

Functions Documented	Total Functions	Coverage %
132	132	100.00%
✓ PASS — Exceeds required threshold of ≥ 80%		

This result exceeds the required threshold (≥ 80%) and satisfies the requested 70–80 compliance band as a minimum acceptance criterion.

2. Measurement Method

Coverage was measured using the following formula:

Code Documentation Coverage

=

(Functions with JSDoc / Total Functions) × 100

The following rules were applied during the scan:

- All JavaScript files under frontend/js were scanned.
- Function declaration and function-expression patterns were counted.
- A function is considered documented only when a valid JSDoc block (`/** ... */`) is directly above it.
- Blank lines between the JSDoc block and the function declaration were tolerated.
- Inline comments (`//`) and block comments (`/* ... */`) that are not valid JSDoc blocks were not counted.
- Anonymous arrow functions assigned to variables were counted when a JSDoc block preceded the assignment.

3. Detailed Results

The table below shows per-file function counts, documented function counts, and individual coverage percentages.

File	Total Functions	Documented	Coverage %
app.js	81	81	100.00%
domain_datasets.js	2	2	100.00%
step2_backend.js	34	34	100.00%
step3_backend.js	13	13	100.00%
step5_backend.js	2	2	100.00%
Overall Total	132	132	100.00%

Overall: 132 / 132 functions documented (100.00%)

4. Priority Function Groups (All Covered)

The following high-priority function groups were explicitly targeted and documented to ensure business-critical readability and maintainability. All groups achieved 100% documentation coverage.

Group	Functions	Status
API	onApplyPreparation, onDetectOutliers, doRealTraining	✓ COVERED
Storage / State	savePreprocessedData, loadPreprocessedData, compressRows, decompressRows	✓ COVERED
Gate / Navigation	gate, showStep, triggerRetrain	✓ COVERED
UI Render / Transform	renderTransformStats, resetTransformStatsPlaceholder, updateHintsWarning	✓ COVERED
Event Wiring	_wireSlider, initStep3UI	✓ COVERED

5. JSDoc Content Standard Applied

Each documented function follows the project documentation standard. The minimum required elements per JSDoc block are:

JSDoc Element	Description	Required
Purpose statement	Plain-language description of what the function does and why.	Always
@param	Input parameter types and their semantic meaning in the application context.	Always
@returns	Return value type and meaning; void noted explicitly where applicable.	Always
@throws	Error types thrown under runtime or API failure conditions.	When relevant
Side effects	Storage writes, DOM mutations, API requests, and global state changes.	When applicable

6. Quality and Runtime Safety Note

JSDoc blocks are comments only. They do not alter runtime behavior, control flow, API payloads, or UI logic.

No functional logic was changed during this documentation pass. The documentation effort was strictly additive — existing function signatures, parameter names, return values, and side-effect behavior were not modified. Lint diagnostics were checked after all updates and no new issues were reported.

7. Conclusion

The HEALTH-AI ML Learning Tool codebase currently achieves 100.00% documentation coverage across all 132 functions in the defined scope (frontend/js/*.js). This result:

- Exceeds the project delivery requirement of $\geq 80\%$ documented functions.
- Satisfies the 70–80 compliance band as a minimum acceptance criterion.
- Covers all five JavaScript source files with individual per-file coverage of 100%.
- Addresses all high-priority API, storage, gate/navigation, UI render, and event-wiring function groups.
- Was achieved without any changes to functional logic — documentation only.

This report is suitable for submission as the Sprint 5 Code Documentation deliverable (SCRUM-106).

Requirement	Threshold	Achieved	Verdict
Code Documentation Coverage	$\geq 80\%$ of functions documented	100.00% (132 / 132)	✓ PASS

